

# Open Banking UK: Read/Write Data API Profile & Security Architecture

AISP, PISP, ASPSP Technical Integration, Consent Lifecycle & Variable Recurring Payments (VRP)

|                   |                                                                  |
|-------------------|------------------------------------------------------------------|
| Standard Setter:  | Open Banking Implementation Entity (OBIE) / Open Banking Limited |
| Specification ID: | OB-RW-API-3.1                                                    |
| Classification:   | Standardized Financial API Architecture                          |
| Verification:     | Integrated in Ariya Sarrafzadeh Production Architecture          |

## 1. Architectural Roles & Topology

Defines interfaces between Account Servicing Payment Service Providers (ASPSPs / Banks), Account Information Service Providers (AISPs), and Payment Initiation Service Providers (PISPs).

## 2. Security Profile: Mutual TLS & OAuth 2.0 FAPI

Every API exchange mandates mutual TLS (mTLS) with Qualified Website Authentication Certificates (QWACs), combined with JSON Web Signatures (JWS detached signatures) for non-repudiation of payment intents.

## 3. Variable Recurring Payments (VRP) & Sweeping

VRP protocols enable continuous, rule-governed fund transfers between accounts owned by the same customer, providing a real-time, low-cost API alternative to traditional Direct Debit mandates.

## 4. Idempotency & Distributed Deduplication

All write requests require unique x-idempotency-key headers. ASPSPs must preserve idempotency state across 24h rolling windows to prevent duplicate execution during network retry spikes.

**Systems Verification Notice:** This authoritative specification establishes the invariant compliance baseline for production fintech switches, settlement reconciliation engines, and double-entry ledger implementations documented across the portfolio.